primal problem

dual problem

Guarantee of equivalence

Another route for algorithms

Motivation for Duality

- Formulation we've seen is called ***primal* problem** (minimization)
- There's an alternative, complementary version, called the ***dual* problem** (maximization)
- Two sides of the same coin!
- Every optimization problem can be associated with a dual, although not always useful, nor equivalent to primal.
- But for LP, they are quite useful:
  - **Guarantee of equivalence** ("strong duality").
  - Dual gives an equivalent but often more interpretable perspective.
  - **Another route for algorithms**!

The Dual Problem

Primal Problem

$$\begin{aligned} \min_{\pi \in \mathbb{R}_+^{n\times m}} \quad & \langle \mathbf{C}, \pi \rangle \\ \text{s.t.} \quad & \pi \mathbf{1} \;\; = \mathbf{a} \\ & \pi^\top \mathbf{1} = \mathbf{b} \end{aligned}$$

Dual Problem